
TRAVAUX DIRIG  S N   1 : Bayes Classifier

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We work in the framework of binary classification. Throughout this tutorial, we consider a random descriptor X taking values in a measurable space $\mathcal{X} \subset \mathbb{R}^d$ ($d \in \mathbb{N}^*$) and a random label Y taking the value 0 or 1. The joint distribution of the vector (X, Y) is denoted by P , and the regression function (posterior probability) is

$$\eta : x \in \mathcal{X} \mapsto \mathbb{P}(Y = 1 \mid X = x) \in [0, 1].$$

EXERCICE 1. We consider $\mathcal{X} = [0, 1]$ and P such that :

- the conditional distribution of X given $Y = 0$ is $P_0 = \mathcal{U}([0, \theta])$, where $\theta \in]0, 1[$,
- the conditional distribution of X given $Y = 1$ is $P_1 = \mathcal{U}([0, 1])$,
- $p = \mathbb{P}(Y = 1) \in]0, 1[$.

For $x \in \mathcal{X}$, give $\eta(x)$ as a function of p and θ . Give the numerical expression for $\theta = 1/2$.

Correction Exercise 1

Since the conditional distributions are uniform and continuous, they admit probability densities. Thus :

$$f_0 : x \in \mathbb{R} \mapsto \frac{1}{\theta} \mathbb{1}_{[0, \theta]}(x), \quad f_1 : x \in \mathbb{R} \mapsto \mathbb{1}_{[0, 1]}(x).$$

Consequently, by the law of total probability, the marginal (unconditional) distribution of X admits the following probability density :

$$f_X : x \in \mathbb{R} \mapsto (1 - p)f_0(x) + pf_1(x) = (1 - p)\frac{1}{\theta} \mathbb{1}_{[0, \theta]}(x) + p\mathbb{1}_{[0, 1]}(x).$$

Now let $x \in [0, 1]$. By Bayes' formula for densities, we have :

$$\eta(x) = \mathbb{P}(Y = 1 \mid X = x) = \frac{pf_1(x)}{f_X(x)} = \frac{p\mathbb{1}_{[0, 1]}(x)}{(1 - p)\frac{1}{\theta} \mathbb{1}_{[0, \theta]}(x) + p\mathbb{1}_{[0, 1]}(x)}.$$

$$\eta(x) = \begin{cases} 1 & \text{if } x \in]\theta, 1] \\ \frac{\theta p}{1 - (1 - \theta)p} & \text{if } x \in [0, \theta] \end{cases}$$

When $\theta = 1/2$, we obtain :

$$\eta(x) = \begin{cases} 1 & \text{if } x \in]\frac{1}{2}, 1] \\ \frac{p}{2 - p} & \text{if } x \in [0, \frac{1}{2}] \end{cases}$$

EXERCICE 2. We consider $\mathcal{X} = \mathbb{R}_+$ and P such that :

- the distribution of X is P_X on $\mathcal{X}$,
- the regression function is $\eta(x) = \frac{x}{x + \theta}$ for every $x \in \mathcal{X}$, where $\theta > 0$ is fixed.

Let h^* denote the Bayes classifier. Make the Bayes classifier explicit in this model. Then show that its 0-1 risk (its probability of error) is

$$L(h^*) = \int_{\mathcal{X}} \min(\eta(x), 1 - \eta(x)) dP_X(x).$$

Compute the Bayes risk when $P_X = \mathcal{U}([0, \alpha\theta])$ with $\alpha > 1$.

Correction Exercise 2

The Bayes classifier for binary labels is given by :

$$h^*(x) = \mathbb{1}_{\{\eta(x) > 1/2\}}.$$

For $x \in \mathcal{X}$, we have :

$$\eta(x) > 1/2 \iff \frac{x}{x + \theta} > \frac{1}{2} \iff 2x > x + \theta \iff x > \theta.$$

Therefore, the Bayes classifier is

$$h^*(x) = \mathbb{1}_{\{x > \theta\}}.$$

We can note that, since the regression function takes values in $[0, 1]$, we also have

$$\eta(x) > 1/2 \iff \eta(x) > 1 - \eta(x).$$

The Bayes 0-1 risk is therefore

$$\begin{aligned} L(h^*) &= \mathbb{E}(\mathbb{1}_{\{Y \neq h^*(X)\}}) = \mathbb{E}(\mathbb{1}_{\{Y=1\}} \mathbb{1}_{\{h^*(X)=0\}} + \mathbb{1}_{\{Y=0\}} \mathbb{1}_{\{h^*(X)=1\}}) \\ &= \mathbb{E}(\mathbb{E}(\mathbb{1}_{\{Y=1\}} \mathbb{1}_{\{h^*(X)=0\}} + \mathbb{1}_{\{Y=0\}} \mathbb{1}_{\{h^*(X)=1\}} \mid X)) \quad (\text{law of total expectation}) \\ &= \mathbb{E}(\eta(X) \mathbb{1}_{\{h^*(X)=0\}} + (1 - \eta(X)) \mathbb{1}_{\{h^*(X)=1\}}) \quad (\text{definition of } \eta \text{ and } h^* \text{ measurable with respect to } X) \\ &= \mathbb{E}(\eta(X) \mathbb{1}_{\{\eta(X) \leq 1 - \eta(X)\}} + (1 - \eta(X)) \mathbb{1}_{\{\eta(X) > 1 - \eta(X)\}}) \\ &= \mathbb{E}(\min(\eta(X), 1 - \eta(X))) = \int_{\mathcal{X}} \min(\eta(x), 1 - \eta(x)) dP_X(x). \end{aligned}$$

When $P_X = \mathcal{U}([0, \alpha\theta])$ with $\alpha > 1$, we have :

$$\begin{aligned}
L(h^*) &= \int_0^{\alpha\theta} \min(\eta(x), 1 - \eta(x)) \frac{1}{\alpha\theta} dx \\
&= \int_0^\theta \frac{x}{x+\theta} \frac{1}{\alpha\theta} dx + \int_\theta^{\alpha\theta} \frac{\theta}{x+\theta} \frac{1}{\alpha\theta} dx \\
&= \frac{1}{\alpha\theta} \left[\int_0^\theta \frac{x}{x+\theta} dx + \int_\theta^{\alpha\theta} \frac{\theta}{x+\theta} dx \right] \\
&= \frac{1}{\alpha\theta} \left[\int_0^\theta 1 - \frac{\theta}{x+\theta} dx + \int_\theta^{\alpha\theta} \frac{\theta}{x+\theta} dx \right] \\
&= \frac{1}{\alpha\theta} \left[\theta - \int_\theta^{2\theta} \frac{\theta}{x} dx + \int_{2\theta}^{(\alpha+1)\theta} \frac{\theta}{x} dx \right] \quad (\text{change of variable } x \mapsto x + \theta) \\
&= \frac{1}{\alpha} [1 - \ln(\theta) + \ln((\alpha+1)\theta) - 2 \ln(2\theta)] \\
&= \frac{1}{\alpha} \ln \left(e \frac{\alpha+1}{4} \right).
\end{aligned}$$

EXERCISE 3. Let $\omega(0), \omega(1) \geq 0$ be weights such that $\omega(0) + \omega(1) = 1$. We consider the weighted classification risk :

$$L_\omega(g) = \mathbb{E} \left(2\omega(Y) \cdot \mathbb{1}_{\{Y \neq g(X)\}} \right), \quad g : \mathcal{X} \rightarrow \{0, 1\}.$$

Give the Bayes classifier and the Bayes risk for this criterion. What is the interest of considering such a criterion?

Correction Exercise 3

The Bayes classifier for the weighted risk is, by definition, the classifier $g^* : \mathcal{X} \rightarrow \{0, 1\}$ that minimizes the weighted risk $L_\omega(g)$. For a classifier $g : \mathcal{X} \rightarrow \{0, 1\}$, define the excess risk with respect to the Bayes classifier g^* by :

$$R_\omega(g) = L_\omega(g) - L_\omega(g^*) \geq 0.$$

For any classifier $g : \mathcal{X} \rightarrow \{0, 1\}$, we have :

$$\begin{aligned}
L_\omega(g) &= \mathbb{E} \left(2\omega(Y) \cdot \mathbb{1}_{\{Y \neq g(X)\}} \right) = 2\mathbb{E} \left(\omega(1) \cdot \mathbb{1}_{\{Y=1\}} \mathbb{1}_{\{g(X)=0\}} + \omega(0) \cdot \mathbb{1}_{\{Y=0\}} \mathbb{1}_{\{g(X)=1\}} \right) \\
&= 2\mathbb{E} \left(\mathbb{E} \left(\omega(1) \cdot \mathbb{1}_{\{Y=1\}} \mathbb{1}_{\{g(X)=0\}} + \omega(0) \cdot \mathbb{1}_{\{Y=0\}} \mathbb{1}_{\{g(X)=1\}} \mid X \right) \right) \quad (\text{law of total expectation}) \\
&= 2\mathbb{E} \left(\mathbb{1}_{\{g(X)=0\}} \cdot \omega(1) \cdot \eta(X) + \mathbb{1}_{\{g(X)=1\}} \cdot \omega(0) \cdot (1 - \eta(X)) \right) \quad (\text{definition of } \eta) \\
&= 2\mathbb{E} \left(\omega(1) \cdot \eta(X) + (\omega(0)(1 - \eta(X)) - \omega(1)\eta(X)) \mathbb{1}_{\{g(X)=1\}} \right) \quad (\mathbb{1}_{\{g(X)=0\}} = 1 - \mathbb{1}_{\{g(X)=1\}}) \\
&= 2\mathbb{E} \left(\omega(1) \cdot \eta(X) + \mathbb{1}_{\{g(X)=1\}} \cdot (\omega(0) - \eta(X)) \right) \quad (\omega(0) + \omega(1) = 1)
\end{aligned}$$

Thus, the excess risk is

$$R_\omega(g) = 2\mathbb{E} \left[(\omega(0) - \eta(X)) (\mathbb{1}_{\{g(X)=1\}} - \mathbb{1}_{\{g^*(X)=1\}}) \right].$$

For every $x \in \mathcal{X}$, we have $(\omega(0) - \eta(x))(\mathbb{1}_{\{g(x)=1\}} - \mathbb{1}_{\{g^*(x)=1\}}) \geq 0$ if and only if :

$$\begin{cases} \omega(0) - \eta(x) \geq 0, \\ \mathbb{1}_{\{g(x)=1\}} - \mathbb{1}_{\{g^*(x)=1\}} \geq 0, \end{cases} \quad \text{or} \quad \begin{cases} \omega(0) - \eta(x) \leq 0, \\ \mathbb{1}_{\{g(x)=1\}} - \mathbb{1}_{\{g^*(x)=1\}} \leq 0. \end{cases}$$

This is satisfied for every g by $g^* : x \in \mathcal{X} \mapsto \mathbb{1}_{\{\eta(x) > \omega(0)\}}$.

The weighted Bayes risk is therefore

$$\begin{aligned}
 L_\omega(g^*) &= 2\mathbb{E} \left(\omega(1) \cdot \eta(X) + (\omega(0) - \eta(X)) \mathbb{1}_{\{\eta(X) > \omega(0)\}} \right) \\
 &= 2\mathbb{E} \left[(\omega(0) + (\omega(1) - 1)\eta(X)) \mathbb{1}_{\{\eta(X) > \omega(0)\}} + \omega(1)\eta(X) \mathbb{1}_{\{\eta(X) \leq \omega(0)\}} \right] \quad \begin{matrix} 1 = \mathbb{1}_{\eta(X) > \omega(0)} + \\ \mathbb{1}_{\eta(X) \leq \omega(0)} \end{matrix} \\
 &= 2\mathbb{E} \left[(1 - \eta(X))\omega(0) \mathbb{1}_{\{\eta(X) > \omega(0)\}} + \omega(1)\eta(X) \mathbb{1}_{\{\eta(X) \leq \omega(0)\}} \right] \\
 &= 2\mathbb{E} \left[\omega(0)(1 - \omega(0) - \eta(X) + \omega(0)) \mathbb{1}_{\{\eta(X) > \omega(0)\}} + \omega(1)(\eta(X) + \omega(0) - \omega(0)) \mathbb{1}_{\{\eta(X) \leq \omega(0)\}} \right] \\
 &= 2\omega(0)\omega(1) + 2\mathbb{E} \left[(\omega(0) - \eta(X)) \mathbb{1}_{\{\eta(X) > \omega(0)\}} \omega(0) + (\eta(X) - \omega(0)) \mathbb{1}_{\{\eta(X) \leq \omega(0)\}} \omega(1) \right] \\
 &= 2\omega(0)\omega(1) - 2\mathbb{E} \left[|\eta(X) - \omega(0)| (\omega(0) \mathbb{1}_{\{\eta(X) > \omega(0)\}} + \omega(1) \mathbb{1}_{\{\eta(X) \leq \omega(0)\}}) \right] \quad |\eta(X) - \omega(0)| < 0
 \end{aligned}$$

We recover the Bayes 0-1 risk when $\omega(0) = \omega(1) = 1/2$:

$$L(g^*) = \frac{1}{2} - \mathbb{E} \left(\left| \eta(X) - \frac{1}{2} \right| \right).$$

Noting that, for every $x \in \mathcal{X}$, $\eta(x) \leq \omega(0)$ if and only if $\omega(1)\eta(x) \leq \omega(0)(1 - \eta(x))$, we obtain :

$$L_\omega(g^*) = \mathbb{E} \left(\min(\omega(1)\eta(X), \omega(0)(1 - \eta(X))) \right).$$

The interest of such a criterion is that it allows classification errors to be weighted differently according to the class. For example, in a medical problem, classifying a sick patient as healthy (a false negative) may be more costly than the reverse (a false positive). By adjusting the weights $\omega(0)$ and $\omega(1)$, we can reflect this asymmetry in the classification risk. Example : a pregnancy test, where false negatives should be avoided, so a higher weight is assigned to class 1.

EXERCISE 4. We consider $X = (T, U, V)$, where T, U, V are i.i.d. real-valued random variables with the standard exponential distribution. Set $Y = \mathbb{1}_{\{T+U+V < \theta\}}$, where $\theta \in \mathbb{R}_+$ is fixed. Compute the Bayes classifier $g^*(T, U)$ when V is unobserved, and then compute its 0-1 risk as a function of θ .

Correction Exercise 4

Let $\eta : (t, u) \in \mathbb{R}_+^2 \mapsto \mathbb{P}(Y = 1 \mid T = t, U = u)$ be the regression function when V is unobserved. By construction,

$$g^*(t, u) : (t, u) \in \mathbb{R}_+^2 \mapsto \mathbb{1}_{\{\eta(t, u) > 1/2\}}.$$

For $t, u \in \mathbb{R}_+$, we have :

$$\begin{aligned}
 \eta(t, u) &= \mathbb{P}(Y = 1 \mid T = t, U = u) = \mathbb{P}(T + U + V < \theta \mid T = t, U = u) \\
 &= \mathbb{P}(V < \theta - t - u) \quad (\text{independence}) \\
 &= F_1(\theta - t - u) = (1 - e^{-(\theta - t - u)}) \mathbb{1}_{\{t + u < \theta\}}. \quad (V \text{ has the standard exponential distribution})
 \end{aligned}$$

Thus $\eta(t, u) > 1/2$ if and only if :

$$\begin{cases} t + u < \theta, \\ 1 - e^{-(\theta - t - u)} > 1/2 \end{cases} \iff \begin{cases} t + u < \theta, \\ t + u < \theta - \ln(2) \end{cases} \iff t + u < \theta - \ln(2).$$

Therefore, the Bayes classifier is $g^*(t, u) = \mathbb{1}_{\{t + u < \theta - \ln(2)\}}$.

The associated Bayes risk is :

$$\begin{aligned}
L(g^*) &= \mathbb{E} (\mathbb{1}_{\{Y \neq g^*(T,U)\}}) = \mathbb{E}(\eta(T,U) \mathbb{1}_{\{g^*(T,U)=0\}} + (1 - \eta(T,U)) \mathbb{1}_{\{g^*(T,U)=1\}}) \\
&= \mathbb{E} (\eta(T,U) \mathbb{1}_{\{T+U \geq \theta - \ln(2)\}} + (1 - \eta(T,U)) \mathbb{1}_{\{T+U < \theta - \ln(2)\}}) \\
&= \mathbb{E} \left((1 - e^{-(\theta - T - U)}) \mathbb{1}_{\{T+U < \theta\}} \mathbb{1}_{\{T+U \geq \theta - \ln(2)\}} + e^{-(\theta - T - U)} \mathbb{1}_{\{T+U < \theta\}} \mathbb{1}_{\{T+U < \theta - \ln(2)\}} \right) \\
&= \mathbb{E} \left((1 - e^{-(\theta - T - U)}) \mathbb{1}_{\{\theta - \ln(2) \leq T+U < \theta\}} + e^{-(\theta - T - U)} \mathbb{1}_{\{T+U < \theta - \ln(2)\}} \right)
\end{aligned}$$

Thus, the Bayes risk can be rewritten as :

$$\begin{aligned}
L(g^*) &= \int_{-\infty}^{+\infty} (1 - e^{-(\theta - x)}) \mathbb{1}_{\{\theta \geq x \geq \theta - \ln(2)\}} f_2(x) dx + \int_{-\infty}^{+\infty} (e^{-(\theta - x)}) \mathbb{1}_{\{x < \theta - \ln(2)\}} f_2(x) dx \\
&= \int_{-\infty}^{+\infty} (1 - e^{-(\theta - x)}) \mathbb{1}_{\{\theta \geq x \geq \theta - \ln(2)\}} x e^{-x} \mathbb{1}_{\{x \geq 0\}} dx + \int_{-\infty}^{+\infty} (e^{-(\theta - x)}) \mathbb{1}_{\{x < \theta - \ln(2)\}} x e^{-x} \mathbb{1}_{\{x \geq 0\}} dx
\end{aligned}$$

If $\theta \geq \ln(2)$, then :

$$\begin{aligned}
L(g^*) &= \int_{\theta - \ln(2)}^{\theta} (1 - e^{-(\theta - x)}) x e^{-x} dx + \int_0^{\theta - \ln(2)} e^{-(\theta - x)} x e^{-x} dx \\
&= \int_{\theta - \ln(2)}^{\theta} x e^{-x} dx - \int_{\theta - \ln(2)}^{\theta} x e^{-\theta} dx + \int_0^{\theta - \ln(2)} x e^{-\theta} dx \\
&= F_2(\theta) - F_2(\theta - \ln(2)) - e^{-\theta} \left[\frac{x^2}{2} \right]_{\theta - \ln(2)}^{\theta} + e^{-\theta} \left[\frac{x^2}{2} \right]_0^{\theta - \ln(2)} \\
&= -(\theta + 1)e^{-\theta} + (\theta - \ln(2) + 1)e^{-(\theta - \ln(2))} - e^{-\theta} \left[\frac{\theta^2}{2} - \frac{(\theta - \ln(2))^2}{2} \right] + e^{-\theta} \left[\frac{(\theta - \ln(2))^2}{2} - 0 \right] \\
&= e^{-\theta} \left[(\theta - \ln(2))^2 - \frac{\theta^2}{2} + \theta - 2 \ln(2) + 1 \right].
\end{aligned}$$

If $\theta < \ln(2)$, then :

$$\begin{aligned}
L(g^*) &= \int_0^{\theta} (1 - e^{-(\theta - x)}) x e^{-x} dx \\
&= \int_0^{\theta} x e^{-x} dx - \int_0^{\theta} e^{-\theta} x dx \\
&= F_2(\theta) - F_2(0) - e^{-\theta} \frac{\theta^2}{2} \\
&= 1 - (\theta + 1)e^{-\theta} - e^{-\theta} \frac{\theta^2}{2}.
\end{aligned}$$

Bibliographic advice

Useful starting points for machine learning are listed below :

- Theoretical and focused on probabilistic aspects : [DGL97]
- Practical and utility-oriented : [HTF13]
- A recent book focused mainly on optimization : [SSBD14] (and, by the same author, on online learning : [SS12])
- Bayesian methods and graphical models : [MB12]

Références

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